

Mini-Project 4

GROUP 3

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Che 396: Senior Design 1

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1. PROJECT CHARTER

Objective:

The objective of this report is to create a thorough safety and environmental hazard assessment for the Vinyl-Chloride process, using the provided process flow diagram and stream table as the foundation for evaluation. The goal is to identify key operational risks, chemical hazards, and potential environmental impacts throughout the process, with a particular focus on the T-100 separation units. The project aims to develop actionable recommendations that improve process safety, regulatory compliance, and environmental performance while ensuring that all risk-mitigation strategies align with best engineering practices and OSHA Process Safety Management guidelines.

Team:

- Anna Cai (Group Leader): Organize Team meetings, enforce deadlines and tasks
- Qussay Suleimann ():
- Miguel Piza (Researcher): Gather Sources, Create & Format Bibliography
- Jazmin Aca (Editor): Ensure that report is formatted properly and flows smoothly.

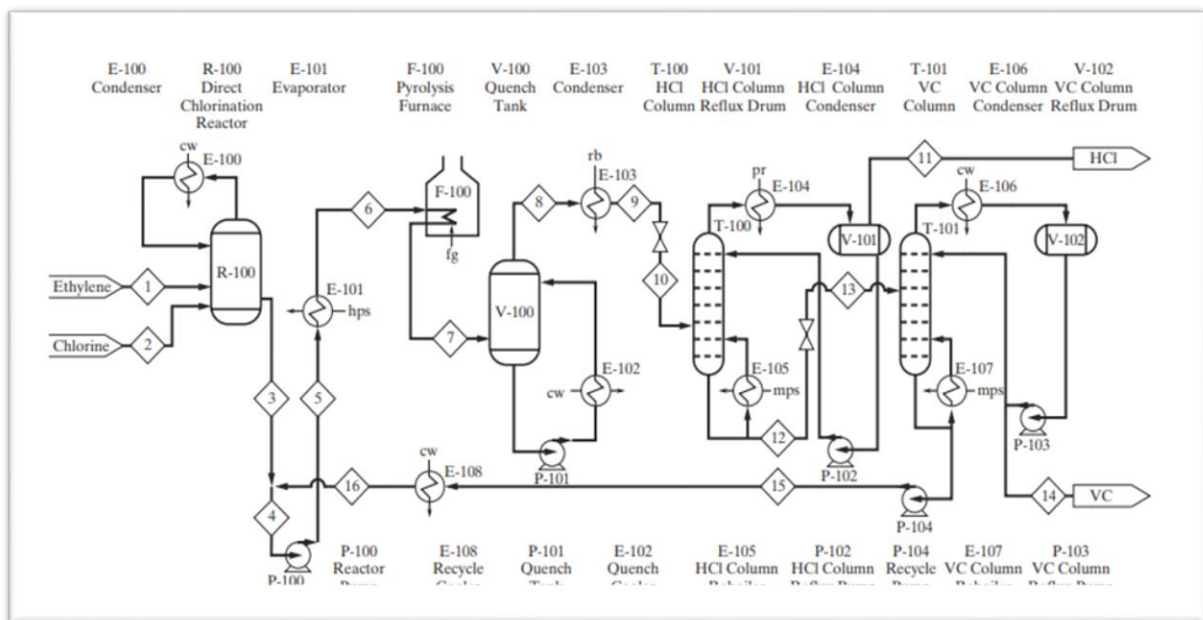
Scope:

- Provide full review of material flows, operating conditions, and hazard prone steps in the vinyl-chloride process, focusing on chemical handling, separation operations, emission pathways, and potential failure modes.
- Evaluation of environmental discharges, safety hazards associated with materials
- Focusing on the T-100 process units as their operation directly influences both process safety and the extent of environmental releases.
- Development of targeted mitigation strategies, HAZOP study, and AI-integration.

Deliverables

- Final Report Including:
 - Project Charter, Design Plan,
 - Comprehensive safety and environmental hazard assessment report for Vinyl Chloride process
 - AI Integration
- Bibliography

2. DESIGN PLAN



Summary of the Process:

The vinyl chloride process consists of a sequence of reaction, separation, and recycle operations that convert ethylene and chlorine into vinyl chloride monomer with high selectivity and minimal raw material waste. The process begins with the direct

chlorination of ethylene in a liquid phase reactor. Ethylene and chlorine react exothermically to form 1,2-dichloroethane, commonly referred to as EDC. Because this reaction generates a large amount of heat and is sensitive to temperature excursions, the reactor is equipped with heat removal systems that maintain a stable operating temperature. Maintaining adequate heat removal helps suppress unwanted side reactions and improves product selectivity. The reactor effluent, which is primarily liquid EDC, is routed to the cracking section where it serves as the primary feed for the second major reaction step.

In the cracking section, the EDC stream is vaporized and heated to a high temperature in a pyrolysis furnace. At elevated temperature, the EDC undergoes an endothermic equilibrium limited reaction to produce vinyl chloride and hydrogen chloride. The reaction conditions are selected to limit single-pass conversion, which helps maintain process operability and reduces the formation of undesirable byproducts. As a result, the furnace effluent contains vinyl chloride, hydrogen chloride, and a significant amount of unconverted EDC that must be recycled. The effluent exits the furnace as a hot gas mixture and requires immediate temperature reduction to stabilize reactive species.

After leaving the furnace, the cracked gas is rapidly cooled and partially condensed by direct contact with cold recycled EDC. This quench step rapidly decreases the temperature of the gas mixture, preventing secondary reactions and uncontrolled polymerization of vinyl chloride. The quench also produces two distinct product streams. The first is a vapor stream enriched in hydrogen chloride and vinyl chloride. The second is a condensed organic phase that contains EDC, along with small amounts of dissolved

products. These streams are directed to downstream separation units for product recovery and purification.

The condensed and vapor phases are processed in a two-stage distillation system that recovers the valuable products and regenerates EDC for recycling. The first column operates as an HCl removal unit. Hydrogen chloride is recovered overhead, while vinyl chloride and EDC are retained in the liquid bottoms product. The column design must address the volatility and corrosiveness of hydrogen chloride and must prevent excessive loss of vinyl chloride in the overhead stream. The bottoms from the HCl column are fed to a second distillation column, which separates vinyl chloride overhead and returns a bottoms stream rich in EDC. The vinyl chloride overhead stream is condensed and collected as the final product, and the EDC-rich bottoms stream is recycled to the cracking section. This recycle loop significantly increases overall conversion, because it recovers unreacted EDC from the separation system and returns it to the reaction network.

The T-100 system consists of the hydrogen chloride removal column, its associated condenser, reboiler, reflux drum, and pumping equipment. This section of the process contains significant quantities of corrosive and toxic materials, primarily hydrogen chloride and chlorinated hydrocarbons, which create inherent risks associated with worker exposure, equipment degradation, and accidental release. A primary safety focus is the prevention of leaks and loss of containment, since even small releases of hydrogen chloride can result in toxic gas clouds, acute health risks, and rapid corrosion of surrounding equipment. Material selection, corrosion monitoring, and appropriate

inspection intervals are therefore fundamental considerations for the safe operation of the T-100 system.

Another central concern is overpressure protection and process upsets. The column is sensitive to disturbances such as excess feed flow, loss of condenser duty, or blocked product outlets, all of which can cause rapid increases in vapor traffic and column pressure. Overpressure can lead to structural failure or forced venting of hazardous vapors, so properly sized pressure relief devices, adequate vent handling systems, and reliable pressure control loops are essential safeguards. Operational disturbances such as loss of cooling or reflux can also destabilize separation performance, resulting in off-spec product, increased HCl carryover to downstream equipment, or flooding within the column. Pump failures, startup transients, and sudden changes in feed composition are particularly relevant operability concerns that require appropriate alarms, interlocks, and shutdown procedures.

Thermal hazards associated with the reboiler and heated equipment must also be addressed. High operating temperatures can accelerate corrosion, promote fouling or polymer formation, and increase the likelihood of tube failure. Loss of heat input or liquid inventory can result in dry-out, while excessive heat input can trigger rapid boiling and pressure excursions. Because vinyl chloride is thermally unstable and can polymerize under uncontrolled conditions, the design and operation of the reboiler must prevent excessive residence times and high localized temperatures. Safe startup and shutdown procedures and inhibitor management are important aspects of risk reduction in this section.

From an environmental standpoint, the T-100 system is a major potential source of air emissions and wastewater contamination. Non-condensable gases from the reflux drum and overhead system may contain hydrogen chloride and trace vinyl chloride, both of which must be captured and treated to meet regulatory standards. Liquid effluents and wash streams can contain acidic and chlorinated compounds, requiring neutralization, phase separation, and possibly stripping or chemical treatment before discharge. Solids such as polymer residues and corrosion products typically meet the definition of hazardous waste and require controlled disposal. In addition, the T-100 system has substantial energy requirements, and the condenser and reboiler impose a significant demand for steam and cooling water, which contributes to the overall environmental footprint of the process. Measures that improve heat integration or reduce utility consumption can therefore provide both environmental and economic benefits.

Overall, the safety and environmental analysis of the T-100 process units should focus on containment of hazardous materials, management of pressure and temperature excursions, prevention of uncontrolled polymerization, and minimization of emissions and waste generation. These considerations influence the selection of construction materials, instrumentation, operational procedures, and environmental control systems that collectively ensure safe, reliable, and compliant operation.

Focus of the Project:

The safety and environmental analysis will concentrate on identifying air emission sources, potential wastewater contamination, and solid by-product generation within the T-100

separation units, which handle volatile hazardous species involved in vinyl-chloride production. Key safety focus areas include evaluation risks related to toxic vinyl chloride exposure, flammability and explosion hazards, thermal excursions, pressure buildup and equipment failure within the T-100 system. The analysis will review applicable OSHA PSM requirements, including permissible exposure limits for highly hazardous chemicals and assess safeguards such as relief systems, containment, ventilation and emergency shutdown protocols.

Assigned Tasks:

- Anna: Process Description
- Qussay: Environmental and Safety Assessment
- Miguel: HAZOP Study, Bibliography
- Jazmin: Project Charter, AI Integration

Gantt Chart:



3. **ENVIRONMENTAL AND SAFETY ASSESSMENT FOR ALL PROCESS UNITS**

This section provides a focused environmental and safety evaluation of the T-100 process units within the vinyl-chloride production system shown in Figure 1. Because these units handle hazardous chlorine-containing intermediates, corrosive HCl streams, and flammable vinyl chloride monomer (VCM), they present several potential environmental and process-safety concerns that must be addressed systematically. This section assesses the key environmental issues associated with gaseous emissions, wastewater generation, and solid-waste residues, followed by a detailed examination of process hazards, regulatory compliance considerations, and recommended safeguards.

Environmental Issues

The T-100 process units within the vinyl-chloride production system present several potential environmental impacts due to the handling of chlorine-containing intermediates, hydrochloric acid streams, and volatile chlorinated hydrocarbons. Air emissions primarily originate from chlorine transfer lines and pump seals feeding R-100, the overhead vents of the T-100 HCl Column (Stream 9) and its reflux drum (Stream 10), and the vent systems associated with the T-101 vinyl chloride column (Stream 11) and the V-102 reflux drum (Stream 14). These vents and fugitive leak points may release chlorine gas, hydrogen chloride vapor, vinyl chloride monomer (VCM), and 1,2-dichloroethane (EDC), all of which are hazardous to human health and the environment. Additional air emissions can arise from E-103 and E-104 condenser vents, where acid mist and residual

chlorinated hydrocarbons may escape if not fully condensed. The overall configuration of the system shown in Figure 1 demonstrates multiple pressurized recycle loops, pumps, and heat-transfer units, each of which represents a potential source of volatile emissions if not properly enclosed.

Wastewater discharges occur primarily in the quench and condensation sections. The V-100 quench tank and its recycle loop (Streams 6 and 8) produce a water stream contaminated with dissolved or entrained EDC, VCM, and acidic components resulting from pyrolysis and chlorination reactions. The bottoms streams of the T-100 HCl Column (Stream 12) and T-101 VC Column (Stream 15) also generate aqueous effluents with residual chlorinated organics and significant acidity due to dissolved hydrogen chloride. Cooling water exchangers such as E-108, E-102, and E-105 may contribute additional trace contamination from leaks or interfacial transfer if equipment integrity is compromised. Without proper treatment, these wastewater streams would pose risks to aquatic ecosystems due to their toxicity, corrosivity, and persistence.

Solid waste generation in the T-100 system primarily results from polymer and coke formation during high-temperature processing. The pyrolysis furnace (F-100), the T-101 VC Column, and associated reboilers (E-105 and E-107) can accumulate polymerized VCM or degraded EDC residues that must be periodically removed during maintenance. These residues are considered hazardous wastes under RCRA regulations because they contain chlorinated hydrocarbons with carcinogenic and mutagenic potential. Additional solid wastes may include spent activated carbon from vapor-recovery systems,

contaminated filters used throughout the plant, and any material removed during furnace decoking or reactor cleaning.

Air-emission mitigation for this process relies on a combination of enclosure, capture, and treatment strategies. All chlorine transfer equipment is designed as a closed system with welded piping where feasible, and pumps handling chlorine or VCM service employ dual mechanical seals or magnetic-drive configurations to minimize fugitive loss. The overhead vents from T-100 and T-101 are routed to a centralized caustic scrubber designed to neutralize hydrogen chloride and absorb chlorine gas. Refrigerated condensers associated with the VC purification system further reduce vapor release by lowering the partial pressure of VCM. For low-concentration vents, a dedicated vapor-recovery unit using activated carbon adsorption is recommended. Similar control strategies are widely used in industrial vinyl-chloride facilities and are consistent with U.S. EPA NESHAP requirements for VCM emissions (EPA, 40 CFR Part 61 Subpart F).

Wastewater mitigation focuses on neutralization and separation of organic phases. Acidic effluents from T-100 and T-101 are first sent to a neutralization basin where the pH is adjusted with sodium hydroxide to remove free hydrogen chloride. Following neutralization, gravity separation or API separators are used to recover EDC or VCM that may phase-separate from water. Water containing dissolved organics undergoes additional treatment through air stripping, steam stripping, or activated-carbon polishing to ensure that volatile chlorinated compounds are removed prior to discharge. These treatment steps reflect common industrial practices for chlorinated-hydrocarbon processes and correspond to the methods described in vinyl chloride environmental

assessments conducted by the European Chemicals Agency (ECHA) and large-scale VCM manufacturers.

Mitigation of solid-waste hazards relies on minimizing polymer formation and establishing proper waste-handling procedures. Temperature control within the pyrolysis furnace and VC Column reduces unwanted polymerization by keeping operating conditions within established limits. Oxygen exclusion is maintained throughout VCM service equipment to prevent oxidative polymer formation. Residues collected from reboilers, column bottoms, and furnace decoking operations are handled as hazardous waste, stored in sealed and labeled containers, and transported to licensed hazardous-waste facilities. Activated-carbon beds used for vapor recovery are regenerated when possible or disposed of according to RCRA requirements. These mitigation techniques mirror those documented in industrial VCM facility safety reviews and academic analyses of chlorinated-hydrocarbon production systems.

Process Safety and Hazard Compliance

This subsection evaluates the primary material and process hazards associated with the T-100 process units in the vinyl-chloride production system shown in Figure 1. The goal is to identify OSHA Process Safety Management (PSM) highly hazardous chemicals present in the system, summarize their relevant toxicological and flammability properties, and assess the equipment-level risks arising from extreme temperatures, pressures, and reactive intermediates. The discussion provides a concise evaluation of the key hazards and the corresponding safeguards incorporated into the design to ensure regulatory compliance and maintain safe operation.

The T-100 process units handle several hazardous chemicals that fall under OSHA PSM requirements, including chlorine, hydrogen chloride, vinyl chloride monomer (VCM), ethylene, and 1,2-dichloroethane (EDC). Chlorine (streams 1, 3) is a highly toxic oxidizer with severe inhalation hazards and is listed explicitly as an OSHA-regulated highly hazardous chemical. Hydrogen chloride in streams 5, 12, and 13 is corrosive and presents acute inhalation toxicity. Vinyl chloride (streams 7, 8, 14) is a flammable gas with established carcinogenicity and is regulated under both OSHA PEL and PSM due to its low exposure limits. Ethylene (streams 1, 2) is a flammable gas that contributes to fire and explosion hazards in feed preparation. EDC (streams 4, 9, 13) is toxic and combustible, with potential exposure risks during handling and distillation operations. These materials require engineering controls that limit worker exposure, prevent releases, and maintain compliance with OSHA and EPA requirements. The permissible exposure limits (PELs) for these substances were obtained directly from their respective Safety Data Sheets. Time-weighted average (TWA) limits are used for all chemicals except hydrogen chloride, which is regulated using a ceiling (C) limit due to its acute inhalation hazard. A summary of the relevant OSHA PEL values is provided below

Table 1: OSHA Permissible Exposure Limits

Chemical	OSHA PEL	Streams
Chlorine	0.5 ppm	1, 3
Hydrogen Chloride	5 ppm	5, 12, 13
Vinyl Chloride	1 ppm	7, 8, 14

Ethylene	NA	1, 2
1,2-Dichloroethane (EDC)	50 ppm	4, 9, 13

The primary process hazards arise from the combination of toxic feeds, reactive intermediates, and high-energy unit operations. The direct chlorination reactor R-100 receives chlorine and ethylene at ambient conditions but undergoes an exothermic reaction that requires cooling to prevent thermal runaway. The feed from R-100 is subsequently heated in the pyrolysis furnace F-100 (stream 5) to approximately 500°C and 26 atm, introducing fire, explosion, and tube-rupture hazards. The T-100 HCl removal column handles hot, corrosive HCl-rich streams (12–13), creating risks related to corrosion, loss of containment, and mechanical degradation. Downstream vinyl-chloride separation towers (T-101, T-102) manage flammable VCM at moderate pressures (24–26 atm) that could polymerize if temperatures rise or inhibitors are depleted. Additionally, the cooling and reflux systems introduce overpressure hazards if heat-exchange failure or blocked flows occur. The key process hazards are summarized below.

Table 2: Summary of Key Process Hazards by Equipment

Equipment	Major Hazard	Relevant Stream Conditions
R-100 Direct Chlorination Reactor	Exothermic reaction, toxic chlorine feed	S1–S3 at ~25°C, 1.5 atm

F-100 Pyrolysis Furnace	Fire/explosion, tube rupture	S5 at 500°C, 26 atm
T-100 HCl Column	Corrosion, HCl exposure	S12–S13
T-101 / T-102 VCM Columns	VCM polymerization, flammability	S7–S8
E-100 Series Heat Exchangers	Overpressure, loss of cooling	Multiple streams

A combination of engineering, administrative, and protective measures ensures compliance with OSHA PSM and reduces the risk of hazardous events. Pressure-relief valves and rupture disks are installed on high-temperature and high-pressure equipment, particularly the furnace, distillation columns, and heat-exchange systems. A safety-instrumented system (SIS) monitors reactor temperature, furnace outlet temperature, and column pressure, providing automated shutdown on abnormal deviations. Corrosion-resistant materials and periodic thickness-testing programs mitigate HCl-induced degradation in the T-100 column and associated piping. Polymerization inhibitors are added to vinyl-chloride streams, and temperature control prevents runaway polymerization in the VCM columns. Secondary containment and leak-detection systems limit chlorine and HCl release impacts. The facility maintains an emergency response plan aligned with OSHA 1910.119 requirements, including evacuation procedures and coordination with local responders.

Overall, the T-100 units present a combination of toxic, corrosive, and flammable hazards driven by the materials handled and the elevated temperature and pressure conditions within the system. The safeguards incorporated into the design significantly reduce the likelihood and consequences of releases, thermal runaway, and polymerization events. These measures ensure compliance with OSHA PSM requirements and create a robust foundation for the subsequent HAZOP evaluation.

4. HAZOP STUDY

Stream 10	Cause	Consequence	Safeguard	Action
No Flow	V-100 not functioning	No feed to columns and therefore no product.	Flow meter monitoring feed.	Maintenance of V-100
Less flow	V-100 not operating functionally	Less feed causing less product.	Flowmeter/control valve.	Maintenance of V-100
More flow	Valves do not function effectively.	Increased reboiler duty and flooding of column	Flowmeter/control valve	Maintenance of V-100

Reverse flow	Too large of pressure in column	No product and build-up of product in V-100	One way valve from S9-S10.	Monitor pressure of column and incoming feed.
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Stream 12	Cause	Consequence	Safeguard	Action
No flow	Tower or E-105 operating at too high temperature	Reduced recovery, purity and no product out from T-101	Temperature sensors on T-100 and E-105	Maintenance of T-100 and E-105 units
Less flow	Tower or E-105 operating at too high temperature	Increased reboiler duty and decreased product.	Temperature sensors on T-100 and E-105	Maintenance of T-100 and E-105 units
More flow	V-101 separating more liquid than desired	Increased reboiler duty or flooding of T-101	Flow rate sensors on S12-S13	Maintenance of V-101
Reverse flow	P-102 not pumping to	Increased liquid content	Pressure sensor on pump and	Monitoring pressure of

	adequate pressure	in T-101, increased reboiler duty or decreased product	one way valve from S12-S13	outgoing stream and column pressure.
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5. AI INTEGRATION

Chat GPT Version 5.1 was given the task of performing the environmental and safety assessment and conducting the HAZOP study for vinyl chloride, this is how it performs.

User Input: Show me how you would perform the safety assessment for the vinyl chloride process.

AI Solution:

1. Define scope and “nodes”

Start from the PFD and Table 1 in the assignment.

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1. **Draw on a printout/PDF of the PFD** and circle “nodes”:

- a. R-100 reactor & quench/evaporator system
- b. T-100 column system (column + reboiler + condenser + reflux drum + pumps)
- c. T-101 VCM column system

- d. Product storage / HCl absorber / incinerator or flare (if shown)
2. Decide where you'll go deep:
- a. **Whole process:** short overview of main hazards & environmental issues.
 - b. **T-100 area:** detailed environmental + safety discussion (because the assignment emphasizes T-100).

In your report this becomes something like:

“The safety and environmental assessment focus on the overall vinyl chloride train, with a detailed analysis of the T-100 de-ethanizer system (column, reboiler, overhead condenser, reflux drum, and associated pumps and piping).”

2. Collect data you need

Before you “assess” anything, gather data in a small internal spreadsheet:

- 1. **From Table 1:** for each stream around T-100 (feeds, overhead, bottom):
 - a. Temperature, pressure
 - b. Phase
 - c. Molar/mass flowrates and compositions (ethylene, 1,2-DCE, VCM, HCl, etc.)
- 2. **From SDS / handbooks (you'd quickly look up):**
 - a. Flammability (flash point, flammable limits in air)
 - b. Acute toxicity (LC50, IDLH, carcinogen status)

- c. Corrosivity, reactivity (polymerization tendency for VCM, etc.)
- 3. Note which chemicals are **OSHA PSM “highly hazardous”** and what their **PEL** values are (these become a table in the report).

This gives you the raw material for both environmental and process-safety sections.

3. Environmental assessment – how I’d do it

I’d literally walk unit-by-unit and ask, for each:

“Where can stuff leave the process and what is in it?”

3.1 Identify emission sources

For each node (especially T-100), list **normal** and **abnormal** release points:

- **Air emissions (VOC / toxic):**
 - Overhead vent from T-100 reflux drum (if vented to flare or atmosphere).
 - Relief valves and rupture disks on T-100 column, reboiler, condenser.
 - Fugitive emissions from pumps, valves, flanges containing VCM, 1,2-DCE, etc.
- **Water / wastewater:**
 - Any condensate collected from overhead systems that could contain dissolved organics (VCM, 1,2-DCE).
 - Equipment drains and sample points going to a sump.
- **Solid / “solid-like” wastes:**

- Any polymerized VCM sludge in reboilers, filters, or knock-out drums.
- Spent desiccants, catalyst residues, spent activated carbon, etc.

You can present this as a table:

Node	Normal emissions	Abnormal emissions	Environmental concern
T-100 overhead system	VOC in vent gas; small organic loss in condensate	Relief valve discharge, condenser tube leak	Air VOCs, toxic VCM and 1,2-DCE, contaminated wastewater

3.2 Characterize pollutants

For each emission source:

1. Use Table 1 compositions to **estimate what chemicals are present** in the overhead/bottom streams.
2. Decide environmental categories:
 - a. **Air:** VOC, hazardous air pollutants (HAPs), greenhouse gases if any.
 - b. **Water:** toxic organics, chlorinated solvents, acidity (HCl).
 - c. **Solids:** hazardous waste (ignitable, toxic, chlorinated).

You don't need full quantitative numbers for a mini-project; a **qualitative severity ranking** (Low/Med/High) is usually enough.

3.3 Propose controls and monitoring

For each significant issue, propose *specific* engineering or design controls, e.g.:

- **T-100 overhead vent gas (VCM + light hydrocarbons):**
 - Route to **flare or thermal oxidizer** instead of direct vent.
 - Consider **vapor recovery** (condenser + carbon bed) to minimize VCM loss.
- **Fugitive VOC emissions:**
 - Use **double mechanical seal pumps**, low-emission valves, and a leak detection and repair (LDAR) program.
- **Contaminated condensate / drains:**
 - Send to **onsite wastewater treatment** (API separator → DAF → biological) with stripping step for chlorinated VOCs.
 - Provide segregated **“dirty” vs “clean” stormwater** systems.

In the report, you’d structure this as:

- 3.1 Identified environmental issues
- 3.2 Recommended controls and design features
- 3.3 Suggested monitoring (stack testing, VOC LDAR, effluent sampling)

4. Process safety / hazard assessment – how I’d do it

Here the question is:

“What could go wrong that hurts people, equipment, or the environment, and what protects us?”

4.1 Inventory hazardous materials

Make a short table with key properties (toxicity + flammability):

- **Vinyl chloride monomer (VCM):** carcinogenic, flammable gas, polymerization risk.
- **1,2-Dichloroethane (1,2-DCE):** toxic, flammable.
- **Ethylene:** flammable gas.
- **Hydrogen chloride (HCl):** highly corrosive, toxic.
- **Chlorine** (if present in upstream steps): toxic, strong oxidizer.

Mark:

- Which are **PSM covered** (e.g., VCM, Cl₂, HCl above threshold quantities).
- Their **permissible exposure limits (PEL)** and IDLH (you’d look these up and cite OSHA / NIOSH).

4.2 Identify main hazards by node

For each major unit, think about **loss of containment, overpressure, wrong conditions**, especially at T-100.

For T-100 column system for example:

- **Overpressure / vacuum:**
 - Loss of condenser duty → pressure rise.
 - Blocked overhead line → pressure rise.
 - Sudden cooling / condensation → column vacuum.
- **Thermal hazards:**
 - Reboiler run-away heat → excessively hot bottoms; potential decomposition or polymerization.
- **Composition / operating deviations:**
 - High light-ends in bottom or product → off-spec and possibly more flammable streams downstream.
 - Too much heavy chlorinated material in overhead → fouling and higher toxicity if vented.
- **Loss of containment:**
 - Column shell leak, pump seal failure, flange leaks, instrument taps, etc.
- **Reactivity / polymerization:**
 - VCM polymerization in trays, reboiler, or piping if inhibitors fail or temperature gets too high.

Do the same high-level pass for R-100, T-101, tanks, etc., but spend the *most* detail on T-100 in the write-up.

4.3 Risk ranking and safeguards

For each hazard scenario, do a qualitative **risk matrix**:

- **Severity:** Catastrophic / Major / Moderate / Minor
- **Likelihood:** Frequent / Occasional / Remote / Improbable

Example (T-100 condenser failure):

- Deviation: “**High pressure** in T-100 due to loss of cooling water to overhead condenser.”
- Consequences: relief valve lift → flammable/toxic gas to flare; if flare not available, potential atmospheric release.
- Severity: Major to Catastrophic.
- Likelihood: Remote (with reliable utilities).
- Existing safeguards:
 - Pressure relief valve sized for full reflux loss.
 - High pressure trip that reduces reboiler duty or shuts feed.
 - Cooling water flow alarms.
- Recommended additional safeguards:
 - Redundant cooling source or emergency cooling.
 - High column pressure interlock that **closes feed and opens vent to flare**.
 - Regular testing of relief devices and trips.

Do this for maybe **5–8 key scenarios** across the process, but with 3–4 belonging to T-100.

You can put these in a table per node:

| Node | Scenario | Consequence | Existing safeguards | Recommended safeguards | Risk rank (before/after) |

5. Connect it with the HAZOP

The assignment has a separate HAZOP section, but practically, the **HAZOP is the structured tool you use to generate the scenarios above.**

For T-100, you'd:

1. Choose **HAZOP nodes**: column feed, column body, overhead system, reboiler, bottoms pump.
2. Apply **guidewords** to key variables (flow, temperature, pressure, level, composition):
 - a. “More flow”, “Less flow”, “No flow”
 - b. “High temperature”, “Low temperature”
 - c. “High pressure”, “Low pressure”
3. For each combination, fill out:
 - a. Deviation
 - b. Possible causes
 - c. Consequences
 - d. Existing safeguards
 - e. Recommendations

Example HAZOP row (overhead system):

- Parameter: Pressure
- Guideword: **High**
- Deviation: High pressure in T-100 overhead and column
- Causes: Loss of condenser cooling, blocked vent line, blocked reflux line
- Consequences: Relief to flare; if flare unavailable, toxic/flammable release (VCM/ethylene); mechanical damage
- Safeguards: Pressure relief valve, high-pressure alarm, automatic reboiler cutback
- Recommendations: Add high-pressure trip on feed; ensure vent routed only to flare; install condenser duty low-flow alarm

The **risk assessment table in section 4.3** then references the most severe HAZOP scenarios.

6. How I'd structure your actual report section

If you're writing the "Environmental and Safety Assessment for T-100 process units" section, I'd structure it like:

3. Environmental and Safety Assessment
 - 3.1 Overview of hazardous materials and operating conditions
 - 3.2 Environmental issues
 - 3.2.1 Air emissions (normal and abnormal)

3.2.2 Wastewater and solid waste

3.2.3 Mitigation measures and monitoring

3.3 Process hazards and safety compliance

3.3.1 Inventory of PSM-covered and hazardous chemicals (with PELs)

3.3.2 Key hazard scenarios for T-100 (overpressure, loss of cooling, leaks, polymerization)

3.3.3 Safeguards and additional design recommendations

3.4 Link to detailed HAZOP results for T-100

If you want, next step I can **fill a sample 1–2 page write-up** for section 3.2 or 3.3 using this structure so you have something concrete to model your own writing on.

User Input:

I need you to perform the HAZOP study and the environmental and safety assessment.

AI Solution:

3. Environmental and Safety Assessment for T-100 Process Units

3.1 Brief description of T-100 system

T-100 is a distillation column that separates light components (e.g., HCl, unreacted ethylene, light hydrocarbons) from heavier chlorinated organics (e.g., 1,2-dichloroethane, vinyl chloride monomer, heavier byproducts). The system includes:

- Feed preheat and feed entry to the column
- Column shell (trays/packing)

- Reboiler and bottoms circuit
- Overhead condenser and reflux drum
- Overhead and bottoms product pumps and associated piping

It handles **toxic, flammable, and corrosive** materials at elevated temperature and moderate pressure, so it is critical both for product quality and for environmental/safety performance.

3.2 Environmental Issues

3.2.1 Air emissions

Potential sources

- **Normal operation**
 - Minor VOC emissions from vents on the **overhead reflux drum** (blanket gas, non-condensables).
 - **Fugitive emissions** from pump seals, valve stems, flanges, and sampling points in services containing:
 - Vinyl chloride monomer (VCM) – a toxic, carcinogenic, flammable gas.[OSHA+1](#)
 - 1,2-dichloroethane (1,2-DCE) – toxic, flammable chlorinated solvent.[CDC+1](#)
 - Hydrogen chloride (HCl) – corrosive, toxic.[OSHA+1](#)
 - Chlorine (if present as impurity or upstream reagent).[OSHA+1](#)

- Use **nitrogen blanketing** and closed vent systems on reflux drums and storage tanks.
- Implement continuous or periodic **VOC monitoring** around the T-100 unit to detect fugitive emissions.

3.2.2 Wastewater discharges

Potential sources

- **Overhead condensate** from T-100, which may contain dissolved VCM, 1,2-DCE, and HCl.
- **Equipment drains and sample lines** used for column reboiler, condenser, and drum drain downs.
- **Cooling water** potentially contaminated by leaks in the overhead condenser.

Environmental concerns

- Chlorinated organics (VCM, 1,2-DCE) are toxic to aquatic life and persistent in the environment.[US EPA+1](#)
- HCl lowers pH and can make effluent corrosive and harmful to receiving waters or biological treatment systems.

Recommended controls

- Route all process-contact condensate and drains to a **segregated, closed wastewater collection system** (not to stormwater).
- Provide **neutralization and VOC stripping** (air stripper or steam stripper) prior to biological treatment.
- Use **double-tubed or plate-and-frame condensers** with leak detection on cooling water outlets to rapidly identify tube leaks.
- Implement routine **effluent monitoring** for chlorinated VOCs, COD, and pH.

3.2.3 Solid and “solid-like” waste

Potential sources

- **Polymerized VCM** and tarry residues in the reboiler, column bottom, filters, and knock-out drums.
- **Spent filters, gaskets, and PPE** contaminated with VCM/1,2-DCE.
- **Spent adsorbents or catalyst** (if used upstream/downstream).

Environmental concerns

- These solids likely meet the definition of **hazardous waste** due to toxicity and, in some cases, flammability.
- Improper disposal could lead to soil and groundwater contamination.

Recommended controls

- Classify all such residues and spent materials as **hazardous waste** under appropriate RCRA codes (e.g., chlorinated solvent wastes).
- Use **closed systems** for reboiler cleaning and residue removal to minimize VOC exposure.
- Contract with **licensed hazardous waste handlers**; maintain manifest and tracking.

3.3 Process Hazards and Safety Compliance

3.3.1 Hazardous materials and OSHA PELs

Key chemicals in T-100 service include:

Chemical	Main hazards	OSHA PEL (U.S.)*
Vinyl chloride (VCM)	Flammable gas; known human carcinogen ; CNS effects; liver damage.	1 ppm 8-h TWA; 5 ppm 15-min STEL (ceiling). <u>OSHA+1</u>
1,2-Dichloroethane (1,2-DCE)	Flammable liquid; toxic, possible human carcinogen; CNS, liver, kidney effects.	50 ppm 8-h TWA; 100 ppm 15-min max; 200 ppm 5-min peak in any 3-h period. <u>NJ.gov+1</u>
Hydrogen chloride (HCl)	Corrosive, toxic gas/vapor; severe respiratory irritation.	5 ppm ceiling (not to be exceeded). <u>OSHA+1</u>

Chlorine (Cl₂)	Strong oxidizer; acute	1 ppm ceiling (general
(if present)	respiratory toxin; corrosive.	industry). <u>OSHA+1</u>

*Check current OSHA and site-specific limits; these values are representative at the time of writing.

Vinyl chloride and chlorine are specifically listed among OSHA Process Safety Management (PSM) highly hazardous chemicals; in addition, large inventories of flammable gases/liquids (e.g., 1,2-DCE, ethylene) may bring T-100 and associated equipment under PSM and Risk Management Program (RMP) coverage when threshold quantities are exceeded.OSHA+1

3.3.2 Process hazards for T-100

Major hazard categories for T-100 include:

- **Overpressure / under-pressure**
 - Loss of condenser duty or blocked overhead line causing **high column pressure**.
 - Condenser over-cooling or rapid condensation causing **column vacuum**.
- **Loss of containment**
 - Leaks from column flanges, manways, reboiler tubes, pump seals, and sample points.
- **Thermal / reactivity hazards**

- Excessive reboiler duty leading to high bottoms temperature, decomposition or polymerization of VCM and heavy chlorinated by-products, and overpressure from rapid boiling.
- **Composition / operability deviations**
 - Poor separation causing light ends in bottoms (off-spec downstream feed) or heavy chlorinated organics in overhead (fouling, high-toxicity vent streams).

3.3.3 Existing and recommended safeguards (summary)

Typical existing safeguards

- Pressure relief valves (PRVs) on the column, reboiler, and condenser, routed to flare.
- Level, temperature, and pressure alarms and trips on column and drums.
- Basic process control loop for column pressure, top/bottom temperature, and reboiler duty.
- Gas detectors in operating areas, PPE and respiratory protection programs.
- Hot-work permitting and lock-out/tag-out procedures.

Recommended additional safeguards for T-100

- **Independent high-pressure and high-temperature trips** that automatically:
 - Cut or reduce feed to T-100.

- Cut reboiler duty.
 - Ensure overhead vent path to flare is open.
- **Vacuum protection** (vacuum breakers or nitrogen make-up) plus low-pressure alarms to prevent shell collapse.
- **Double isolation and bleed** arrangements for critical maintenance points (reboiler, overhead condenser).
- **Polymerization control:**
 - Maintain minimum inhibitor concentration where appropriate.
 - Avoid stagnant regions in reboiler and low-flow piping.
 - Set limits on maximum bottoms temperature and residence time.
- **Improved leak detection:**
 - Fixed VOC and HCl/Cl₂ detectors near pumps, column base, and overhead drum.
 - Routine LDAR surveys with action limits.
- **Emergency response:**
 - Clearly defined **emergency venting and evacuation procedures** for VCM/Cl₂/HCl releases.
 - On-site spill kits and neutralizing agents for HCl.
 - Coordination with local emergency responders for toxic gas releases.

4. HAZOP Study – T-100 Process Units

Below is a structured HAZOP focused on key T-100 nodes. You can copy these tables into your report and adjust wording/format.

4.1 Nodes considered

1. **Node 1 – Feed to T-100** (from upstream reactor/column + feed preheat)
2. **Node 2 – Column body (trays/packing)**
3. **Node 3 – Reboiler and bottoms circuit**
4. **Node 4 – Overhead condenser and reflux drum**
5. **Node 5 – Overhead product line / vent to flare**

Node 1 – Feed to T-100

Parameter	Guideword / Deviation	Possible causes	Consequences	Existing / assumed safeguards	Recommendations
Flow	No flow to column	Upstream pump trip;	Column inventory decreases;	Feed flow alarm;	Add low feed-flow trip to reduce reboiler duty; ensure orderly shutdown logic
		upstream unit shutdown;	separation stops; possible	interlock from	
		blocked feed line;	loss of pressure	upstream	

		closed	control; upset		
		valve	to downstream		
			units		
		Control	Flooding of		Size control
		valve	column;	Flow	valve
	More flow	failure	entrainment;	control	appropriately;
Flow	than	open;	poor	loop;	add high-flow
	design	operator	separation;	operator	alarm and
		setpoint	potential high	monitorin	interlock to cap
		error	column	g	feed rate
			pressure		
			Extra		High/low feed
		Feed	vaporization in		temperature
Temperat	Low feed	preheater	column; higher	Temperat	alarm;
ure	temperatu	failure; low	condenser	ure	investigate
	re	upstream	duty; off-	control on	minimum feed
		temperature	design	preheater	temperature for
			operation		safe operation
	High feed	Preheater		Preheater	Add
Temperat	temperatu	runaway;	Excess vapor	temperatu	independent
ure	re	wrong	load at top;	re control;	high feed-
		heat-	high column	column	temperature

		exchanger	pressure; risk	pressure	trip that
		service	of relief	control	reduces feed or bypasses preheater
	Off-spec				
	composition	Upstream	T-100 design		Tighten
	(e.g.,	reactor	separation may		upstream
	more light	upset;	be inadequate;	Upstream	control; add
Composition	ends or	wrong	off-spec	quality	online analyzers
on	more	recycle	overhead/bottoms; higher	control;	(e.g., GC) on
	heavy	routing;	toxicity and	lab	feed to alarm on
	chlorinate	contaminated	flammability	analysis	composition
	d	on	of vent streams		swings
	organics)				

Node 2 – Column body (trays/packing)

Parameter	Guideword / Deviation	Possible causes	Consequences	Existing / assumed safeguards	Recommendations
Pressure	High pressure in column	Condenser duty loss;	Lifting of PRV; flammable/t	PRV to flare; pressure	Add independent high-pressure trip to reduce reboiler duty

		blocked	oxic	indicator &	and close feed; verify
		overhead	discharge	control;	PRV sizing for worst
		line;	to flare or	high-	credible case
		incorrect	atmosphere	pressure	
		setpoint;	; potential	alarm	
		high feed	mechanical		
		temperatu	damage		
		re			
		Over-			
		condensat	Column		
		ion;	shell		
		sudden	collapse;	Vacuum	Ensure vacuum
		cooling;	air ingress	breaker or	protection is installed
Pressure	Low	shutdown	and	nitrogen	and sized; add low-
	pressure /	with hot	formation	padding (if	pressure alarm; avoid
	vacuum	column;	of	installed)	rapid cooling during
		steam	flammable		shutdown
		condensat	mixtures		
		ion			
Tempera	High	Excess	Thermal	Temperatu	Add high-temperature
ture	temperat	reboiler	decomposit	re control;	trips that reduce
profile	ures (e.g.,	duty; loss	ion or	alarms	reboiler duty; evaluate

	at bottom	of	polymerizat		safe temperature
	or mid-	internal	ion of		limits based on
	section)	reflux	chlorinated		decomposition/polym
		from low	species;		erization data
		condenser	fouling;		
		duty;	high		
		control	pressure		
		failure	from		
			excessive		
			vaporizatio		
			n		
			Poor		
			separation;		
			entrainment		
	High	Overfeed;	; liquid		Implement tray ΔP
	liquid	vapor rate	carry-over;	Normal	and flooding
		too high;	off-spec	instrument	indicators; alarm on
	level in	tray	products;	ation	abnormal pressure
	trays	damage;	possible		drop patterns
	(flooding)	fouling	liquid		
			slugging to		
			overhead		
Level					

Mechanical integrity	Tray/packing damage	Corrosion	Use materials
		Reduced capacity; maldistribution; local hot spots	compatible with chlorinated streams and HCl; implement cleaning plans to control polymer deposition
		Periodic inspection during turnarounds	

Node 3 – Reboiler and bottoms circuit

Parameter	Guideword / Deviation	Possible causes	Consequences	Existing / assumed safeguards	Recommendations
Flow	No circulation (no flow through reboiler)	Pump trip; blocked line; valve closed; vapor lock	Local overheating; decomposition of heavy organics; rapid pressure rise; potential tube rupture	Reboiler flow indicator; low-flow alarm (if any)	Add low reboiler-flow interlock to cut steam duty; design for fail-closed steam to

					reboiler on
					trip
					Independent
		Excess			high-
		steam		Temperat	temperature
		pressure		ure	trip on
Temperat	High		Decomposition/polyme	controlle	bottoms or
	reboiler	; control	rization; increased non-	r;	reboiler outlet;
ure	temperat	valve	condensable formation;	pressure	define safe
	ure	failure;	fouling and plugging	control	temperature
		wrong		on steam	limit in
		steam			operating
		supply			procedures
		Excessi			Add low-level
		ve			trip that shuts
	Low	draw-	Loss of thermosyphon	Level	bottoms pump
	bottoms	off;	flow; potential dry-fire	control	and reduces
Level	level	pump	in reboiler; tube	loop	steam; audible
		cavitati	overheating		alarm
		on; leak			

Level	High bottoms level	Level			Add high-level interlock
		controll			to cut feed and
		er	Flooding of lower	Level	reboiler duty;
		failure;	trays; poor separation;	indicator	provide
		blocked	potential carry-over to	& alarm	overflow path
		draw	reboiler; thermal stress		to a safe,
		line			contained
					sump
					Use corrosion-
					resistant
		Corrosi			metallurgy;
		on by			planned
		HCl and	Release of hot, toxic,	Relief	thickness
		chlorina	chlorinated fluid to	valve;	monitoring;
Contain	Leak / tube rupture	ted	atmosphere or into	containm	double-tubed
		organic	steam condensate; risk	ent dike	design if
ment		s;	of exposure and	(if any)	failure risk is
		erosion;	environmental release		high; route
		thermal			condensate to
		cycling			closed system
					for monitoring

Node 4 – Overhead condenser and reflux drum

Parameter	Guideword / Deviation	Possible causes	Consequences	Existing / assumed safeguards	Recommendations
Temperature	High overhead temperature (insufficient condensation)	Loss of cooling water; fouled condenser; high feed/reboiler duty	High column pressure; PRV lift; increased VCM/1,2-DCE emission to flare/vent; off-spec overhead	Pressure control; high-pressure alarm; PRV	Add low cooling-water flow and high condenser outlet temperature alarms; high-pressure trip that cuts reboiler duty and feed
	Low overhead temperature (over-condensation)	Excess cooling; cold weather; control failure	Potential vacuum formation; accumulation of light ends in drum; risk	Temperature/pressure control	Implement low-pressure alarm and vacuum protection; define minimum

			of freezing		allowable
			or hydrate		overhead
			formation		temperature
			(if		
			applicable)		
		High	Liquid		
		condensed	carry-over		Add high-high
		flow;	to vent		level trip to
		failed	system;		stop feed
Level in		level	possible	Level control	and/or reboiler;
reflux	High level	controller;	liquid to	loop; high-level	ensure vent
drum		blocked	flare; loss	alarm	system can
		overhead	of		handle two-
		product	inventory		phase flow
		line	control		safely
		Low	Loss of		Low-level
		condensati	reflux →		interlock to
Level in	Low level	on; pump	poor		stop feed and
reflux	(no	trip;	separation;	Level alarm	reboiler duty;
drum	reflux)	leakage;	rapid		automatic
		incorrect	change in		switchover to
			column		standby reflux

Containment	Condenser tube leak	level	temperature	Cooling-water monitoring (pH, TOC)	pump if installed
		setpoint	and pressure		
			Cross-contamination of cooling		Use corrosion-resistant materials; install
		Corrosion from HCl/Cl ₂ ; erosion	water with VCM/1,2-DCE; environmental release via cooling tower or discharge		hydrocarbon/VOC detection on cooling-water outfall; provide rapid isolation and rerouting to a closed drain

Node 5 – Overhead product and vent system

Parameter	Guideword		Consequences	Existing /	
	rd / Deviation	Possible causes		assumed safeguards	Recommendations

Flow	No flow to flare/vent	Blocked vent line; isolation valves closed; frozen or plugged header	Column overpressure; failure to relieve; catastrophic rupture	PRV (but may still be blocked if downstream m blocked)	Strict management of change for any vent modifications; lock-open status and car-seals for critical isolation valves; regular testing of flare/vent system
					Re-validate flare capacity with current inventory/composition; consider staged or secondary relief to dedicated scrubber/thermal oxidizer for
Flow	More flow than design (large relief)	Fire, external heating; multiple PRVs lifting simultaneously	Overloading flare; high radiation; unburnt toxic/flammbale gas release	Flare design for worst-case scenario	

					chlorinated
					streams
					Assess need for
	More	Upstream	Higher		wet caustic
	toxic/	upset;	HCl/Cl ₂ /VC	Materials	scrubber
Compositi	more	heavy	M content in	selection;	upstream of flare
on	chlorinat	components	vent, causing	current	for acid gases;
	ed than	carried	increased	flare/scrub	upgrade
	expected	overhead	corrosion	ber design	metallurgy if
			and toxicity		necessary

4.2 Summary of HAZOP findings

From the T-100 HAZOP, the **highest-risk scenarios** typically include:

- **High column pressure** due to condenser failure or blocked overhead line → potential large toxic/flammable release via relief devices.
- **High reboiler temperature / loss of circulation** → decomposition, polymerization, and overpressure.
- **Condenser tube leaks** → hidden contamination of cooling water and air emissions via cooling tower.
- **Loss of reflux and level control in the reflux drum** → severe upset of separation and vent system.

The proposed safeguards focus on:

- Adding or strengthening **instrumented protection layers** (independent trips on pressure, temperature, and levels).
- Ensuring **robust relief and vent routing** to flare / scrubber systems.
- Improving **materials selection and corrosion management** for chlorinated, acidic service.
- Implementing strong **environmental controls and monitoring** (wastewater treatment, LDAR, VOC & acid gas monitoring).

If you want, I can next:

- Turn this into a **clean Word-style outline** with numbered subsections, or
- Add a **very short “executive summary” paragraph** you can paste near the front of the report.

Reflection:

While AI tools such as ChatGPT can provide helpful starting points, explanations, and draft analyses, they are not suitable for completing this assignment on their own. Process safety and environmental hazard evaluations require precise alignment with what we have learned in class, the specific methodologies taught in the textbook, and the detailed expectations of chemical engineering practice, areas where AI models can drift,

generalize, or fabricate information. For example, AI may misinterpret process conditions, overlook unit-specific hazards, or provide generic safety recommendations that do not match the exact vinyl-chloride process presented in this course. Additionally, AI cannot independently verify regulatory requirements, such as OSHA PSM thresholds or permissible exposure limits, and may produce outdated or incomplete values unless guided carefully. Because of these limitations, any output generated by AI must be critically reviewed, cross-checked with class notes, and corrected using validated engineering principles. Ultimately, AI can support but not replace the engineering judgment, accuracy, and process-specific reasoning required to deliver a reliable and academically sound hazard assessment.

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